

## Hardware & Networking

### Data Center Watch: Token Spend Grows, While GPU Pricing Remains Firm, and Memory Inflation Starts to Cool

*In the Data Center Watch report, we outline key data points relevant to AI infrastructure demand, including LLM token volumes and prices, GPU rental prices, and memory costs. Taken together, the latest data suggest positive trends in the AI infrastructure demand environment, including continued expansion in LLM usage and firm pricing, both of which are driving further increases in total spend. In July, GPU rental price growth for B200s accelerated vs. declined for H100s, possibly reflecting a shift in demand toward the latest generation of GPUs. Meanwhile, the growth in contract prices for both DRAM and NAND is expected to moderate substantially to ~15% q/q in 3Q26 relative to high-double digits growth in 2Q26, given long-term agreements, weaker consumer demand, and 1H26 pull-forward. Key highlights from Data Center Watch report:*

- **LLM growth trends moderate on OpenRouter for the month of July following robust June.** LLM growth trends on OpenRouter in July moderated sequentially following a strong June: Token Volume tracked to +18% m/m and 21x y/y, Pricing tracked to -3% m/m and +2% y/y, and overall Spending tracked to +7% m/m and 11x y/y. Interestingly, while closed-weight models represented the minority volume share (29%), they continued to dominate spend (80%); however, open-weight models are encroaching on spend (20% share in July vs. 12% in June vs. 10% in May), with both pricing (VWAP +36% m/m) and volume (+26% m/m) growing rapidly as the models get closer to frontier capabilities. Notably, Kimi K3 has broken into the top five models by spend, a category that has been historically exclusive to closed-weight models.
- **A100 and H100 GPU rental price growth rates fall, while B200 price growth accelerates.** GPU rental price sequential growth trends were mixed in July for Nvidia GPUs with regards to non-hyperscaler capacity. B200s led the m/m gains, posting the only sequential increase among the three, while A100s expanded at a more measured pace and H100s declined modestly. Pricing for B200s was ~2.1x that of H100s, and H100 pricing was ~1.6x that of A100s, with the former ratio rising sequentially and the latter ratio continuing its decline for the fourth month in a row.
- **Growth in contract prices for both DRAM and NAND is expected to moderate substantially in 3Q26.** Per TrendForce, DRAM and NAND contract prices are expected to grow +16% q/q and +13% q/q in 3Q26, respectively, representing a substantial deceleration from +56% q/q in 2Q26 for DRAM (vs. +96% q/q in 1Q26) and +58% q/q for NAND (vs. +88% q/q in 1Q26). TrendForce attributed expectations for the moderation in contract price growth for DRAM and NAND to higher base effects, weakening consumer demand, long-term agreements with CSPs, and elevated buyer inventories following 1H26 pull-forwards.

### LLM Token Tracker

*Our LLM Token Tracker captures a single week of usage and pricing data each month, providing a monthly snapshot or sampling of trends across ~300 LLMs listed on OpenRouter, an aggregation platform that routes API requests to models*

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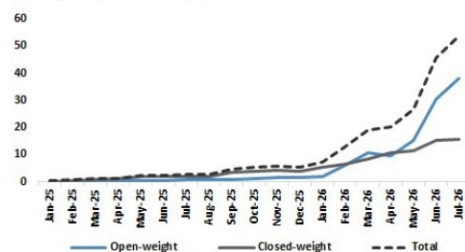
through a single endpoint. Keep in mind, OpenRouter skews towards developer, startup, and agentic-coding traffic and excludes first-party API volume (direct OpenAI/Anthropic, hyperscaler-hosted endpoints).

Key takeaways:

- Token volume growth moderated in the month of July, following a meaningful acceleration in June.** Token volume growth moderated in July, expanding +18% m/m (vs. +70% in June and +33% in May) and 21x y/y (vs. 20x in June and 12x in May). Volumes for closed-weight models (OpenAI, Anthropic, etc.), which represented 29% of the total (vs. 33% in June and 43% in May), also saw growth decelerate in July, expanding +3% m/m (vs. +34% in June and +7% in May) and 8x y/y (vs. 8x in June and 6x in May). Meanwhile, volumes for open-weight models (DeepSeek, Moonshot AI, etc.), also saw growth decelerate in July, expanding +26% m/m (vs. +97% in June and +63% in May) and 63x y/y (vs. 74x in June and 46x in May).
- Token pricing modestly up year-over-year in July.** While average token pricing fell -3% m/m in July (vs. +7% in June and +5% in May), year-over-year declines stopped, with pricing expanding +2% y/y (vs. -5% in June and -21% in May). We believe this modest improvement was led by strong pricing trends for open-weight models, as evident in volume-weighted average pricing, which rose +36% m/m for open-weight models (vs. +12% in June and -49% in May) and +74% y/y (vs. +25% in June and +24% in May) vs. -5% m/m in July for closed-weight models (vs. +23% in June and +22% in May) and +18% y/y (vs. +76% in June and +38% in May).
- Token spending also slowed in July, following strong growth in June.** Token spending growth slowed in July, expanding +7% m/m (vs. +70% in June and +23% in May) and 11x y/y (vs. 16x in June and 9x in May). Spending for closed-weight models, which represented 80% of the total (vs. 88% in June and 90% in May), saw growth slow in July, declining -2% m/m (vs. +65% in June and +30% in May) and 9x y/y (vs. 14x in June and 8x in May). By comparison, spending for open-weight models also saw growth decelerate in July, increasing +71% m/m (vs. +120% in June and -18% in May) and 109x y/y (vs. 92x in June and 57x in May).
- Kimi K3 breaks into top 5 models by spend.** The top five models in July based on token usage were: 1) MiMo v2.5; 2) DeepSeek v4 Flash; 3) GLM 5.2; 4) DeepSeek v4 Pro; and 5) MiniMax M3, accounting for 50% of total token volume. However, on a spending basis, the top five models in July were: 1) Claude Opus 4.8; 2) Claude Opus 4.7; 3) Fable 5; 4) Kimi K3; and 5) GPT-5.6 Sol, accounting for 54% of total token spend.

Figure 1: Token Volumes

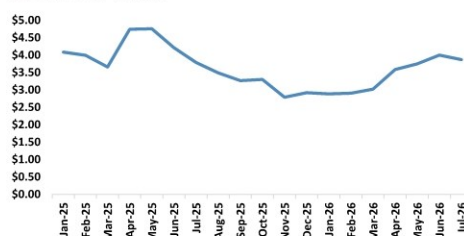
Weekly Tokens in Trillion



Source: OpenRouter, J.P. Morgan Estimates

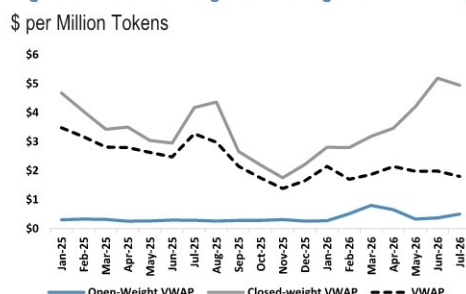
Figure 2: Average Token Pricing

\$ per Million Tokens



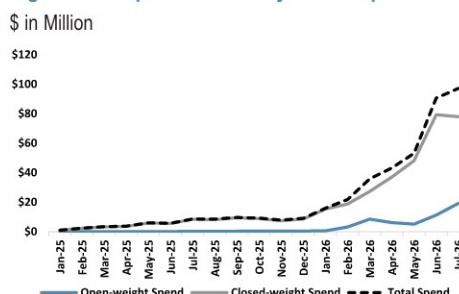
Source: OpenRouter, J.P. Morgan Estimates

Figure 3: Volume-Weighted Average Token Pricing



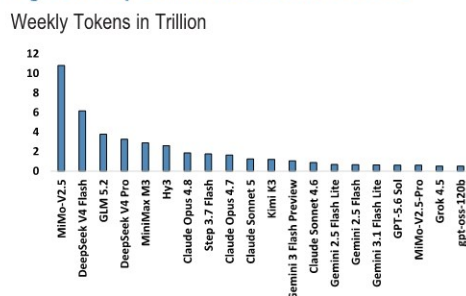
Source: OpenRouter, J.P. Morgan Estimates

Figure 4: Snapshot of Weekly Token Spend



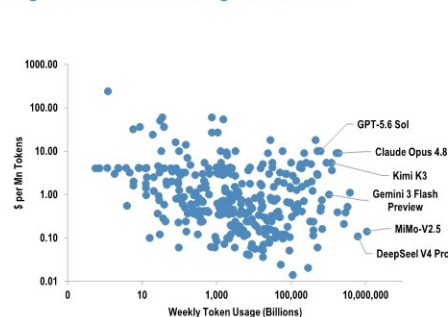
Source: OpenRouter, J.P. Morgan Estimates

Figure 5: Top 20 Models Based on Volume



Source: OpenRouter, J.P. Morgan Estimates

Figure 6: Token Pricing vs. Volume



Source: OpenRouter, J.P. Morgan Estimates

## GPU Rental Price Tracker

Key takeaways based on Bloomberg indices:

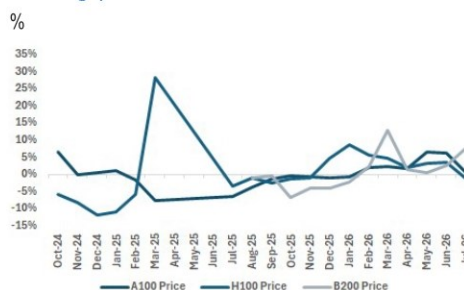
- **A100 rental prices are up modestly m/m in July.** A100 rental prices tracked to \$1.66/GPU-hour on average in July, up +0.9% m/m (vs. +6.4% m/m in June and +6.6% m/m in May), representing a sixth straight month of sequential growth.
- **H100 rental prices tracking down m/m in July; H100-to-A100 ratio ticks lower.** H100 rental prices tracked to \$2.68/GPU-hour on average during July, a -1.1% m/m decrease (vs. +3.5% m/m in June and +3.2% m/m in May), marking the first sequential decline after seven straight months of sequential growth. The H100-to-A100 price ratio tracked to 1.63x on average in July (vs. 1.66x in June and 1.71x in May).
- **B200 rental price growth accelerated m/m in July; B200-to-H100 ratio rebounds.** B200 rental prices tracked to \$5.72/GPU-hour on average in July, up +7.3% m/m (vs. +2.7% m/m in June and +0.4% m/m in May), with the m/m increase much larger than the sequential moves for the H100 and A100. The B200-to-H100 price ratio tracked to 2.13x on average in July (vs. 1.97x in June, 1.98x in May, and 2.58x around index launch in Sep-25).

Figure 7: Avg. Monthly Rental Prices (Non-Hyperscaler)



Source: Bloomberg Finance L.P.

Figure 8: Avg. Monthly Rental Prices (M/M %Change)



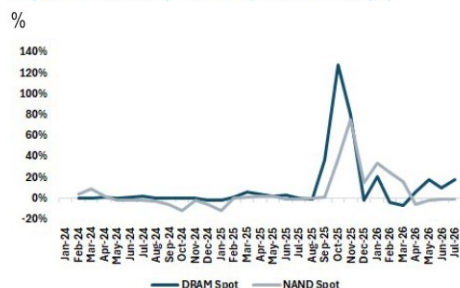
Source: Bloomberg Finance L.P.

## Memory Price Tracker

Key takeaways based on Bloomberg data:

- **DRAM spot prices continued to rise in July, with DDR5 16Gb up more than 8x y/y.** DDR5 16Gb spot prices reached \$50.83 in July, rising +17% m/m (vs. \$43.14 in June and \$39.24 in May) and +890% y/y (vs. \$5.13 in July-25), marking a fourth consecutive m/m increase following modest sequential declines in February and March.
- **NAND spot prices decline further m/m in July; NAND 1Tb still up more than 4x y/y.** NAND 1Tb spot prices tracked to \$26.90 in July, a decline of -0.5% m/m (vs. \$27.03 in June and \$27.12 in May) and an increase of +417% y/y (vs. \$5.25 in July-25), marking a fourth consecutive month of modest m/m declines following the strong run-up through 1Q26.
- **DRAM and NAND contract prices expanded +56% and +58% q/q in 2Q26; Contract prices to expand into 3Q26E as well, albeit at a moderated pace.** As per TrendForce, DRAM contract prices expanded +56% q/q in 2Q26 (vs. +96% q/q in 1Q26), while NAND contract prices increased +58% q/q (vs. +88% q/q in 1Q26). For 3Q26E, DRAM and NAND contract prices are expected to expand +16% q/q and +13% q/q, respectively.

Figure 9: Memory Prices (M/M %Change)



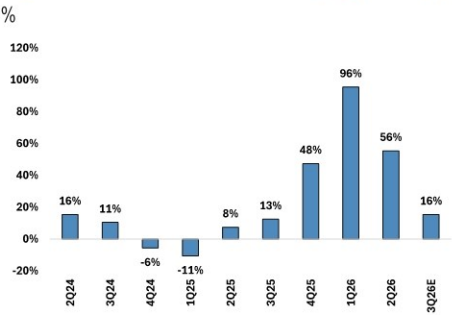
Source: Bloomberg Finance L.P.

Figure 10: Memory Prices (Y/Y % Change)



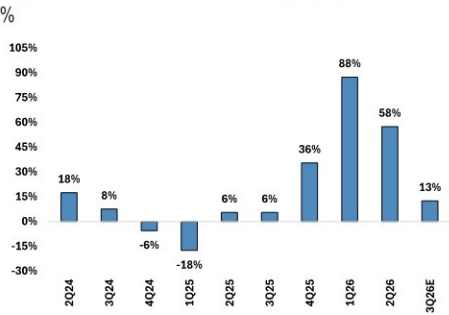
Source: Bloomberg Finance L.P.

Figure 11: DRAM Contract Price Q/Q growth (%)



Source: TrendForce.

Figure 12: NAND Contract Price Q/Q growth (%)



Source: TrendForce.

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